

Module 2: Data Preprocessing & Normalization

Data Cleaning, Chi-Square Correlation, Min-Max, Z-Score & PCA Reduction

Module II: Data Preprocessing & Processing — 6 Topics Syllabus

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| 1. Data Cleaning (Missing Values, Noise Binning, Regression, Outliers) | 2. Data Integration (Entity Identification, Chi-Square, Pearson, Spearman, Covariance) |
| 3. Data Transformation (Min-Max, Z-Score, Decimal Scaling, Box-Cox, Log) | 4. Data Reduction (PCA Derivation, SVD, Wavelets, Numerosity, Sampling, Compression) |
| 5. Data Discretization (Binning, Histograms, Cluster, ChiMerge Step-by-Step) | 6. Concept Hierarchy Generation (Categorical & Numeric 3–4–5 Rule) |

Topic 1: Data Cleaning (Noise, Missing Values & Outliers)

Real-world industrial datasets are inherently "dirty" — incomplete (lacking attribute values or containing only aggregate data), noisy (containing errors, outliers, or transmission corruptions), and inconsistent (containing discrepancies in codes, names, or measurement units).

Data cleaning routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies.

1. Advanced Strategies for Handling Missing Attribute Values:

- Ignore the Tuple:** Typically done when the target class label is missing in supervised classification tasks. This method is inefficient when the percentage of missing values per attribute varies considerably across records.
- Fill in the Missing Value Manually:** High accuracy but extraordinarily time-consuming, expensive, and infeasible for large multi-gigabyte databases.
- Use a Global Constant:** Replace all missing attribute values with a fixed label such as "Unknown" or a sentinel value ($-\infty$). Simple, but may lead the mining algorithm to infer false correlations among tuples sharing the global constant.
- Use Attribute Mean or Median:**
 - Use the **Arithmetic Mean** for symmetric (Gaussian) continuous distributions.
 - Use the **Median** for highly skewed distributions (e.g., household income or real estate prices).
- Use the Attribute Mean/Median for All Samples of the Same Class:** If classifying credit risk, fill missing salary with the mean salary of customers sharing the same credit classification label.
- Use the Most Probable Value (Statistical Imputation):** Determine the missing value using linear regression, Bayesian networks, inference-based tools, or decision tree induction. Considered the most statistically rigorous approach.
- K-Nearest Neighbor (KNN) Imputation:** Identifies the k most similar complete records based on distance metrics (e.g., Euclidean distance) and imputes the weighted average of the neighbors' values:

$$\hat{x}_i = \frac{\sum_{j \in N_k(i)} \frac{1}{d(x_i, x_j)} x_j}{\sum_{j \in N_k(i)} \frac{1}{d(x_i, x_j)}}$$

- Multiple Imputation by Chained Equations (MICE):** A series of iterative regression models where each missing variable is modeled conditionally on other variables in the dataset over multiple imputation cycles.

2. Smoothing Noisy Data via Binning Techniques:

Binning methods smooth sorted data values by consulting their neighborhood (local values around each point):

Step-by-Step Solved Problem 1: Binning Smoothing Methods

Sorted Data for Price Attribute: 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34 ($n = 12$).

Step 1: Partition into 3 Equal-Frequency (Equal-Depth) Bins (Depth = 4 items/bin):

- Bin 1: [4, 8, 9, 15]
- Bin 2: [21, 21, 24, 25]
- Bin 3: [26, 28, 29, 34]

Step 2: Smoothing by Bin Means:

- Mean(Bin 1) = $\frac{4+8+9+15}{4} = \frac{36}{4} = 9 \implies$ Bin 1: [9, 9, 9, 9]
- Mean(Bin 2) = $\frac{21+21+24+25}{4} = \frac{91}{4} = 22.75 \implies$ Bin 2: [23, 23, 23, 23]
- Mean(Bin 3) = $\frac{26+28+29+34}{4} = \frac{117}{4} = 29.25 \implies$ Bin 3: [29, 29, 29, 29]

Step 3: Smoothing by Bin Boundaries (Min and Max of Each Bin):

- Bin 1 (Min = 4, Max = 15): $8 \rightarrow 4, 9 \rightarrow 15 \implies$ [4, 4, 15, 15]
- Bin 2 (Min = 21, Max = 25): $24 \rightarrow 25 \implies$ [21, 21, 25, 25]
- Bin 3 (Min = 26, Max = 34): $28 \rightarrow 26, 29 \rightarrow 26 \implies$ [26, 26, 26, 34]

3. Regression & Outlier Clustering for Smoothing:

- **Linear Regression:** Fits data to a straight line $y = wx + b$. Where slope $w = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$ and intercept $b = \bar{y} - w\bar{x}$. Values are replaced by their predicted regression coordinates.
- **Multiple Linear Regression:** Fits multidimensional attributes to a hyper-plane $y = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$.
- **Cluster Analysis for Outlier Detection:** Similar values are organized into clusters. Values falling outside major cluster boundaries (e.g. distance $> 3\sigma$ from centroid) are flagged as outliers.
- **Local Outlier Factor (LOF):** Measures local density deviation of a given object with respect to its surrounding k -nearest neighbors. An object with density substantially lower than its neighbors is identified as a local outlier.

Topic 2: Data Integration & Redundancy Correlation Analysis

Data Integration combines data from multiple heterogeneous sources into a coherent store. Key challenges include:

- **Entity Identification Problem:** Matching real-world entities across databases (e.g., ensuring `customer\_id` in DB1 matches `cust\_number` in DB2 using schema metadata).
- **Redundancy & Correlation:** An attribute may be redundant if it can be derived from another attribute. Redundancy leads to dimensional bloat, increased storage overhead, and algorithm bias.

1. Chi-Square (χ^2) Correlation Test for Nominal Attributes:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad \text{where } E_{ij} = \frac{\text{row_total}_i \times \text{col_total}_j}{N}$$

Step-by-Step Solved Problem 2: Chi-Square (χ^2) Contingency Table Test

Contingency Table (Gender vs. Fiction Book Preference):

Gender	Prefers Fiction	Prefers Non-Fiction	Row Total
Male	250 (Observed O_{11})	200 (Observed O_{12})	450
Female	50 (Observed O_{21})	100 (Observed O_{22})	150
Col Total	300	300	$N = 600$

Step 1: Compute Expected Frequencies (E_{ij}):

- $E_{11} = \frac{450 \times 300}{600} = 225$, $E_{12} = \frac{450 \times 300}{600} = 225$
- $E_{21} = \frac{150 \times 300}{600} = 75$, $E_{22} = \frac{150 \times 300}{600} = 75$

Step 2: Compute χ^2 Statistic:

$$\chi^2 = \frac{(250 - 225)^2}{225} + \frac{(200 - 225)^2}{225} + \frac{(50 - 75)^2}{75} + \frac{(100 - 75)^2}{75} = \frac{625}{225} + \frac{625}{225} + \frac{625}{75} + \frac{625}{75} = 2.78 + 2.78 + 8.33 + 8.33 = 22.22$$

Step 3: Hypothesis Testing: Degrees of freedom $\nu = (r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$. The critical value of χ^2 at significance $\alpha = 0.001$ is 10.83. Since $22.22 > 10.83$, the hypothesis of independence is strongly rejected $\Rightarrow$ **Gender and Book Preference are statistically correlated!**

2. Correlation Coefficient (Pearson's r) for Numeric Data:

$$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{(n - 1)s_A s_B} = \frac{\sum_{i=1}^n (a_i b_i) - n\bar{A}\bar{B}}{(n - 1)s_A s_B}$$

- $r > 0$: A and B are **positively correlated** (values of A increase as B increases).
- $r = 0$: A and B are **independent / uncorrelated**.
- $r < 0$: A and B are **negatively correlated** (as A increases, B decreases).

3. Spearman's Rank Correlation Coefficient (ρ):

Used for ordinal attributes or monotonic non-linear numeric data:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Where $d_i = \text{rank}(a_i) - \text{rank}(b_i)$ is the difference between ranks of tuple i .

Step-by-Step Solved Problem: Spearman's Rank Correlation

Given Student Exam Ranks in Data Mining (X) and Machine Learning (Y):

Student	Rank X	Rank Y	$d_i = X - Y$	d_i^2
S1	1	2	-1	1
S2	2	1	1	1
S3	3	4	-1	1
S4	4	3	1	1
S5	5	5	0	0
Total	-	-	-	$\sum d_i^2 = 4$

Calculation ($n = 5$):

$$\rho = 1 - \frac{6 \times 4}{5(25 - 1)} = 1 - \frac{24}{120} = 1 - 0.20 = \mathbf{0.80}$$

Strong positive rank correlation ($\rho = 0.80$) between Data Mining and Machine Learning performance!

4. Covariance Between Numeric Attributes:

$$\text{Cov}(A, B) = E[(A - \bar{A})(B - \bar{B})] = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n - 1}$$

Note: $r_{A,B} = \frac{\text{Cov}(A,B)}{s_A s_B}$. If $\text{Cov}(A, B) > 0$, both attributes vary in the same direction.

Topic 3: Data Transformation (Normalization Formulations)

Data transformation converts the data into forms appropriate for mining, preventing attributes with initially large ranges (e.g., `Salary` in thousands) from dominating attributes with smaller ranges (e.g., `Age` in tens).

Method	Mathematical Formulation	Key Advantages & Tradeoffs
1. Min-Max Normalization	$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$	Preserves exact linear relationships among data values; maps values to $[0, 1]$; encounters "out-of-bounds" error on future data exceeding range.
2. Z-Score Normalization (Zero-Mean)	$v' = \frac{v - \mu_A}{\sigma_A}$	Transforms data to have mean $\mu = 0$ and standard deviation $\sigma = 1$; extraordinarily robust when min and max are unknown or extreme outliers exist.
3. Decimal Scaling	$v' = \frac{v}{10^j} \quad \text{where } j \text{ is smallest integer such that } \max(v') < 1$	Fast bit-shift operation; normalizes by moving decimal point.
4. Box-Cox Power Transform	$y^{(\lambda)} = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln(x) & \text{if } \lambda = 0 \end{cases}$	Parametric power transformation used to stabilize variance and normalize non-Gaussian positive distributions.
5. Log Transformation	$v' = \log_{10}(v) \quad \text{or} \quad \ln(v + 1)$	Compresses large values and stretches small values; ideal for right-skewed revenue/income curves.

Step-by-Step Solved Problem 3: Normalization Calculations

Given: Attribute `Income` with $\min = \$12,000$, $\max = \$98,000$, $\mu = \$54,000$, $\sigma = \$16,000$. Transform value $v = \$73,600$.

1. **Min-Max Normalization to Range** $[0.0, 1.0]$:

$$v' = \frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0.0) + 0.0 = \frac{61,600}{86,000} = \mathbf{0.716}$$

2. **Z-Score Normalization:**

$$v' = \frac{73,600 - 54,000}{16,000} = \frac{19,600}{16,000} = \mathbf{1.225}$$

3. **Normalization by Decimal Scaling** ($\max = 98,000 \implies j = 5$ since $98,000/10^5 = 0.98 < 1$):

$$v' = \frac{73,600}{10^5} = \mathbf{0.736}$$

Topic 4: Data Reduction (PCA Mathematical Derivation & Numerosity)

Complex data analysis and mining on massive datasets may take an impractical amount of time. **Data Reduction** techniques obtain a reduced representation of the dataset that is much smaller in volume yet closely maintains the integrity of the original data.

1. Complete Step-by-Step Mathematical Derivation of PCA:

Worked Problem 4: Complete PCA Step-by-Step Matrix Derivation

Given 4 2D Data Points: $x_1 = (2, 4)$, $x_2 = (4, 6)$, $x_3 = (6, 8)$, $x_4 = (8, 10)$ ($n = 4$, $d = 2$).

1. Step 1: Compute Attribute Means:

$$\mu_1 = \frac{2 + 4 + 6 + 8}{4} = 5.0, \quad \mu_2 = \frac{4 + 6 + 8 + 10}{4} = 7.0$$

2. Step 2: Construct Zero-Mean Centered Matrix (X):

$$X = \begin{bmatrix} 2-5 & 4-7 \\ 4-5 & 6-7 \\ 6-5 & 8-7 \\ 8-5 & 10-7 \end{bmatrix} = \begin{bmatrix} -3 & -3 \\ -1 & -1 \\ 1 & 1 \\ 3 & 3 \end{bmatrix}$$

3. Step 3: Calculate Covariance Matrix ($C = \frac{1}{n-1} X^T X$):

$$C = \frac{1}{3} \begin{bmatrix} -3 & -1 & 1 & 3 \\ -3 & -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} -3 & -3 \\ -1 & -1 \\ 1 & 1 \\ 3 & 3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 20 & 20 \\ 20 & 20 \end{bmatrix} = \begin{bmatrix} 6.67 & 6.67 \\ 6.67 & 6.67 \end{bmatrix}$$

4. Step 4: Solve for Eigenvalues ($\det(C - \lambda I) = 0$):

$$\det \begin{bmatrix} 6.67 - \lambda & 6.67 \\ 6.67 & 6.67 - \lambda \end{bmatrix} = (6.67 - \lambda)^2 - (6.67)^2 = \lambda^2 - 13.33\lambda = 0$$
$$\lambda(\lambda - 13.33) = 0 \implies \lambda_1 = 13.33, \quad \lambda_2 = 0.0$$

5. Step 5: Compute Eigenvector for Principal Component $\lambda_1 = 13.33$:

$$\begin{bmatrix} 6.67 - 13.33 & 6.67 \\ 6.67 & 6.67 - 13.33 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix} = \begin{bmatrix} -6.67 & 6.67 \\ 6.67 & -6.67 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies e_{11} = e_{12}$$

$$\text{Normalized unit eigenvector: } e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}.$$

6. Step 6: Proportion of Total Variance Explained:

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{13.33}{13.33 + 0.0} = 100\%$$

The dataset can be compressed from 2 dimensions down to 1 dimension with **zero information loss!**

2. Discrete Wavelet Transforms (DWT) & Haar Decomposition:

DWT transforms a numeric vector X of length $N = 2^m$ into wavelet coefficients via hierarchical subband decomposition.

Step-by-Step Solved Problem: Haar Wavelet Decomposition

Given 8-Element Signal: $S = [2, 2, 4, 4, 6, 6, 8, 8]$.

1. **Level 1 Averages (Approximation):** $\left[\frac{2+2}{2}, \frac{4+4}{2}, \frac{6+6}{2}, \frac{8+8}{2}\right] = [2, 4, 6, 8]$.

Level 1 Differences (Detail): $\left[\frac{2-2}{2}, \frac{4-4}{2}, \frac{6-6}{2}, \frac{8-8}{2}\right] = [0, 0, 0, 0]$.

2. **Level 2 Averages:** $\left[\frac{2+4}{2}, \frac{6+8}{2}\right] = [3, 7]$.

Level 2 Differences: $\left[\frac{2-4}{2}, \frac{6-8}{2}\right] = [-1, -1]$.

3. **Level 3 Average (Overall Mean):** $\frac{3+7}{2} = 5$.

Level 3 Difference: $\frac{3-7}{2} = -2$.

4. **Full Haar Wavelet Transformed Vector:** $[5, -2, -1, -1, 0, 0, 0, 0]$. Zero detail coefficients can be truncated with zero reconstruction loss!

Topics 5 – 6: Data Discretization & Concept Hierarchy Generation

Data Discretization reduces the number of values for a given continuous attribute by dividing the range of the attribute into distinct intervals.

The ChiMerge (χ^2 Merge) Algorithm:

ChiMerge is a supervised, bottom-up discretization method based on the χ^2 test.

1. Sort continuous attribute values in ascending order. Each distinct value forms an initial interval.
2. For each pair of adjacent intervals, compute the χ^2 statistic based on class distribution.
3. Find the pair of adjacent intervals with the smallest χ^2 value (lowest difference in class labels).
4. If the minimum $\chi^2 < \text{threshold}$, merge the two intervals.
5. Repeat steps 2–4 until all adjacent intervals have $\chi^2 \geq \text{threshold}$ or number of intervals reaches maximum limit.

Concept Hierarchy Generation by the 3–4–5 Rule:

The **3–4–5 Rule** segments numeric intervals naturally into human-friendly chunks:

- If an interval covers 3, 6, 7, or 9 distinct units at the most significant digit, partition the range into **3 equal-width intervals** (3 for 3, 6, 9; 3 for 7 as 2 – 3 – 2).
- If an interval covers 2, 4, or 8 distinct units, partition the range into **4 equal-width intervals**.
- If an interval covers 1, 5, or 10 distinct units, partition the range into **5 equal-width intervals**.

Q1. Explain the step-by-step mathematical procedure of Principal Component Analysis (PCA) for data reduction. (10 Marks)

Answer: 1. **Standardize Input Data:** Given n data vectors of dimension d , normalize each attribute to zero mean: $x_{ij} \leftarrow x_{ij} - \bar{x}_j$.

2. **Calculate Covariance Matrix:** Compute $d \times d$ covariance matrix $C = \frac{1}{n-1} X^T X$, where $C_{jk} = \text{cov}(X_j, X_k)$.

3. **Compute Eigenvalues & Eigenvectors:** Solve characteristic equation $\det(C - \lambda I) = 0$ to find d eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$ and orthonormal eigenvectors $e_1, e_2, \dots, e_d$.

4. **Select Top k Components:** Choose k eigenvectors corresponding to largest eigenvalues such that cumulative variance explained $\frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^d \lambda_j} \geq 0.95$.

5. **Project Data:** Transform original $n \times d$ matrix X onto k -dimensional orthogonal subspace: $Y = X \cdot E_k$, reducing dimensionality from d to k while preserving maximum variance.

Q2. Differentiate between Stratified Sampling and Simple Random Sampling with respect to data reduction. (6 Marks)

Answer: In Simple Random Sampling (SRS), every tuple in dataset D has an equal probability of selection ($1/N$). If data contains rare skewed classes (e.g., credit card fraud occurring in 0.1% of cases), SRS may completely omit all positive fraud instances in the sample.

In **Stratified Sampling**, the dataset is partitioned into mutually exclusive non-overlapping strata based on a key attribute (e.g., class label). Random samples are drawn independently from each stratum proportional to its representation, guaranteeing that rare classes are preserved in the reduced dataset.

Q3. Why is data preprocessing considered the most time-consuming phase in the KDD pipeline? (6 Marks)

Answer: Real-world operational data is collected from heterogeneous systems without unified validation. Studies show data preprocessing consumes 60–80% of total project time because it must resolve missing values, eliminate noise, reconcile disparate primary/foreign keys across schemas, convert incompatible data formats, remove redundant correlated attributes, and scale continuous numerical ranges without destroying underlying statistical relationships.

Q4. Compare Min-Max Normalization and Z-Score Normalization. When should each be chosen? (6 Marks)

Answer: 1. **Min-Max Normalization** bounds data strictly into $[0, 1]$, preserving exact original relative distances. Choose Min-Max when algorithms require inputs in a bounded range (e.g., Neural Network sigmoid activations, Image pixel values).

2. **Z-Score Normalization** transforms data to $\mu = 0, \sigma = 1$ without upper/lower bounds. Choose Z-Score when the absolute minimum and maximum are unknown or when severe outliers exist, as Min-Max would compress all normal data into a microscopic interval.